INCH-POUND
MIL-DTL-24308/3J
w/AMENDMENT 4
12 June 2020
SUPERSEDING
MIL-DTL-24308/3J
w/AMENDMENT 3
2 May 2017

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRIC, RECTANGULAR, MINIATURE, POLARIZED SHELL, RACK AND PANEL, PLUG, PIN CONTACTS, GENERAL PURPOSE, CLASS D AND G, SOLDER TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24308.

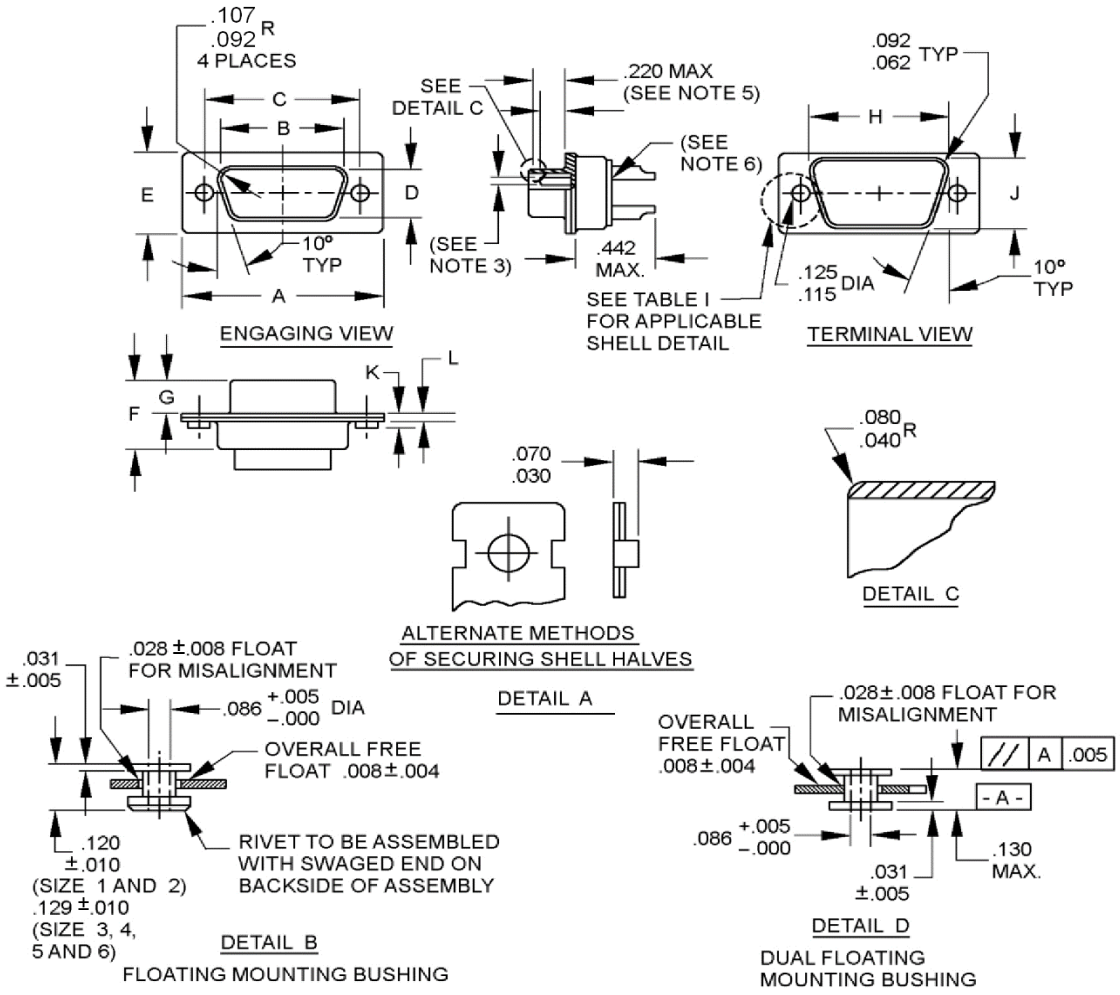


FIGURE 1. Shell, plug, solder type.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:03Z
Check the source to verify that this is the current version before use.

Detail (See note 7)	Shell size	Dimensions										
		A	B	C	D	E	F	G	H	J	K	L
A	1	1.228 (31.19) 1.198 (30.43)	.671 (17.04) .661 (16.79)	.989 (25.12) .979 (24.87)	.334 (8.48) .324 (8.23)	.509 (12.93) .479 (12.17)	.432 (10.97) .412 (10.46)	.238 (6.05) .229 (5.82)	.769 (19.53) .749 (19.02)	.432 (10.97) .412 (10.46)	.060 (1.52) .035 (.89)	.040 (1.02) .020 (.51)
	2	1.556 (39.52) 1.526 (38.76)	.999 (25.37) .989 (25.12)	1.317 (33.45) 1.307 (33.20)	.334 (8.48) .324 (8.23)	.509 (12.93) .479 (12.17)	.432 (10.97) .412 (10.46)	.238 (6.05) .229 (5.82)	1.093 (27.76) 1.073 (27.25)	.432 (10.97) .412 (10.46)	.060 (1.52) .035 (.89)	.040 (1.02) .020 (.51)
	3	2.103 (53.42) 2.073 (52.65)	1.539 (39.09) 1.529 (38.84)	1.857 (47.17) 1.847 (46.91)	.334 (8.48) .324 (8.23)	.509 (12.93) .479 (12.17)	.436 (11.07) .416 (10.57)	.236 (5.99) .224 (5.69)	1.635 (41.53) 1.615 (41.02)	.432 (10.97) .412 (10.46)	.070 (1.78) .050 (1.27)	.049 (1.24) .029 (.74)
	4	2.744 (69.70) 2.714 (68.94)	2.187 (55.55) 2.177 (55.30)	2.505 (63.63) 2.495 (63.37)	.334 (8.48) .324 (8.23)	.509 (12.93) .479 (12.17)	.436 (11.07) .416 (10.57)	.236 (5.99) .224 (5.69)	2.282 (57.96) 2.262 (57.45)	.432 (10.97) .412 (10.46)	.070 (1.78) .050 (1.27)	.049 (1.24) .029 (.74)
	5	2.650 (67.31) 2.620 (66.55)	2.084 (52.93) 2.074 (52.68)	2.411 (61.24) 2.401 (60.99)	.446 (11.33) .436 (11.07)	.620 (15.75) .590 (14.99)	.436 (11.07) .416 (10.57)	.236 (5.99) .224 (5.69)	2.188 (55.58) 2.168 (55.07)	.544 (13.82) .524 (13.31)	.070 (1.78) .050 (1.27)	.049 (1.24) .029 (.74)
	6	2.744 (69.70) 2.714 (68.94)	2.217 (56.31) 2.207 (56.06)	2.505 (63.63) 2.495 (63.37)	.508 (12.90) .498 (12.65)	.683 (17.35) .653 (16.59)	.436 (11.07) .416 (10.57)	.236 (5.99) .224 (5.69)	2.312 (58.72) 2.292 (58.22)	.606 (15.39) .586 (14.88)	.070 (1.78) .050 (1.27)	.049 (1.24) .029 (.74)
B (See note 8)	1	Same as detail A, shell size 1 except in accordance with detail B.										
	2	Same as detail A, shell size 2 except in accordance with detail B.										
	3	Same as detail A, shell size 3 except in accordance with detail B.										
	4	Same as detail A, shell size 4 except in accordance with detail B.										
	5	Same as detail A, shell size 5 except in accordance with detail B.										
	6	Same as detail A, shell size 6 except in accordance with detail B.										
D	1	Same as detail A, shell size 1 except in accordance with detail D.										
	2	Same as detail A, shell size 2 except in accordance with detail D.										
	3	Same as detail A, shell size 3 except in accordance with detail D.										
	4	Same as detail A, shell size 4 except in accordance with detail D.										
	5	Same as detail A, shell size 5 except in accordance with detail D.										
	6	Same as detail A, shell size 6 except in accordance with detail D.										

FIGURE 1. Shell, plug, solder type - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:03Z
Check the source to verify that this is the current version before use.

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Pin diameter .041/.039 (1.04/0.99mm) for arrangements A-1-1, A-2-1, A-3-1, A-4-1, A-5-1. Pin diameter .0305/.0295 (0.775/0.749 mm) for arrangements A-1-2, A-2-2, A-3-2, A-4-2, A-5-2 and A-6-1.
4. Copyright notice. All information disclosed in this military specification which is of may be copyrighted by ITT Cannon Electric is reproduced herein with the express permission of the copyright owner.
5. Dimensions are .150 (3.81mm) minimum full pin diameter extension (sizes 3 through 6) and .159 (4.04mm) minimum full pin diameter extension (sizes 1 and 2).
6. Indentation on the ends of the termination side of the shell, for the purpose of securing the insulator in the shell, is permitted. This may result in the “F” dimension being undersized in these areas. However, the “F” dimension shall be maintained for the remainder of the connector.
7. See table II for supersession data.
8. Detail B is inactive for new design, use detail D.

Inches	mm	Inches	mm	Inches	mm
.004	0.10	.062	1.57	.115	2.92
.005	0.13	.065	1.65	.120	3.05
.008	0.20	.070	1.78	.125	3.18
.010	0.25	.086	2.18	.129	3.28
.030	0.76	.092	2.34	.130	3.30
.031	0.79	.102	2.59	.220	5.59
.040	1.02	.106	2.69	.422	10.72

FIGURE 1. Shell, plug, solder type – Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:03Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

The connectors shall conform to figure 1 and the specification Part or Identifying Numbers (PIN) indicated in table I.

TABLE I. Design standards. 1/

PIN Class G	PIN Class D	Shell size	Shell Detail (see figure 1)	Insert arrangement reference number 3/	Contact PIN 4/
M24308/3-1	M24308/3-23	1	A	A-1-1	B2
M24308/3-2	M24308/3-24	2	A	A-2-1	B2
M24308/3-3	M24308/3-25	3	A	A-3-1	B2
M24308/3-4	M24308/3-26	4	A	A-4-1	B2
M24308/3-5	M24308/3-27	5	A	A-5-1	B2
2/ M24308/3-12	M24308/3-34	1	B	A-1-1	B2
M24308/3-13	M24308/3-35	2	B	A-2-1	B2
M24308/3-14	M24308/3-36	3	B	A-3-1	B2
M24308/3-15	M24308/3-37	4	B	A-4-1	B2
M24308/3-16	M24308/3-38	5	B	A-5-1	B2
2/ M24308/3-17	M24308/3-39	1	D	A-1-1	B2
M24308/3-18	M24308/3-40	2	D	A-2-1	B2
M24308/3-19	M24308/3-41	3	D	A-3-1	B2
M24308/3-20	M24308/3-42	4	D	A-4-1	B2
M24308/3-21	M24308/3-43	5	D	A-5-1	B2

1/ These connectors mate with connectors specified in MIL-DTL-24308/1, MIL-DTL-24308/2, and MIL-DTL-24308/23.
2/ PINs M24308/3-12 through M24308/3-16 and M24308/3-34 through M24308/3-38 are inactive for new design, see M24308/3-17 through M24308/3-21 and M24308/3-39 through M24308/3-43.
3/ For insert arrangement see MIL-DTL-24308, Appendix A.
4/ For contact see MIL-DTL-24308, Appendix B.

PIN: Consists of the letter M, the basic number of the specification sheet, and a dash number compiled from the code.

PIN example: M24308/3 -35 F

Specification sheet number

Data sheet number from table

Finish: For class G:

A = Pure electrodeposited aluminum
F = Cadmium
K = Zinc nickel
P = Passivated stainless steel
T = Nickel Fluorocarbon Polymer
Z = Zinc

For class D: Class D connectors have a nickel finish, no suffix is required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:03Z
Check the source to verify that this is the current version before use.

Supersession data: See table II.

TABLE II. Shell supersession data.

Shell detail (see figure 1)	Superseded MS sheet
Detail A	MS18268-1 through -5
Detail B	MS18268-7 through -11
Detail D	MS18268-13 through -17

Amendment notations. The margins of this specification are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-24308, this document references the following:

MIL-DTL-24308/1
MIL-DTL-24308/2
MIL-DTL-24308/23

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA
DLA - CC

Preparing activity:
DLA - CC

(Project: 5935-2020-064)

Review activities:
Army - AT, CR4, MI
Navy - AS, CG, MC, SH

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:03Z
Check the source to verify that this is the current version before use.